ARES-Planner: Autonomous Resource, Energy & Science Observation Constraint Scheduler on Hera LEON3 Bare-Metal Core

EUROPEAN SPACE AGENCY (ESA) – OPEN SPACE INNOVATION PLATFORM (OSIP)

Call for Ideas: Autonomous Software Experiments on Hera | Category 3 – Spacecraft Operations Optimization

Proposal Identifier:	ESA-OSIP-HERA-2026-RDAS-OPS-ARES-PLANNER	**Target Processor:**	GR712RC Dual-Core LEON3 (SPARC V8 @ 50 MHz)
Proposing Entity:	radixal s.r.o. (Purkynova 649/127, 612 00 Brno, Czech Republic)	**Execution Core:**	Core 1 Isolated Bare-Metal Sandbox (No OS, 0 malloc)
Leadership Triad:	Bc. Viktor Lostak (PI), Ing. Petr Slepicka, Mgr. David Riedl	**Applicable Standards:**	ECSS-E-ST-40C Category D | MISRA-C:2012 Zero-Heap
Primary Science Target:	Didymos / Dimorphos Binary Asteroid System	**In-Flight Slot:**	Extended Mission Campaign (August 2027, 4 Weeks)

EXECUTIVE SUMMARY & KEY IN-FLIGHT BUDGETS:
ARES-Planner is a lightweight, deterministic onboard observation scheduler executing in Core 1 bare-metal C. It autonomously orchestrates multi-payload observation sequences (AFC, PALT, TIRI, HyperScout-H) by solving a bounded Constraint-Satisfaction Problem (CSP) directly on board.

- CPU Utilization: 4.8% @ 50 MHz SPARC V8 (Peak WCET: 1.4 s per 24-hour planning epoch)
- RAM Footprint: 42.8 kB Static RAM | < 12.0 kB Stack (Zero dynamic allocation / No malloc)
- Scientific Return Gain: +35% increase in valid science observation targets per orbit
- Constraint Safety: Formal mathematical guarantee against battery/thermal over-draw
- Telemetry Emission: PUS Science Packets (APID 0x483) containing optimized timeline plans.

Abstract & Table of Contents

Abstract: Multi-instrument asteroid observation requires dynamic scheduling. The ARES-Planner experiment deploys a deterministic integer Branch-and-Bound Constraint-Satisfaction Problem (CSP) solver on Core 1 bare-metal C. Evaluating battery voltage, memory capacity, and orbit geometries at 4.8% CPU load, ARES-Planner delivers a +35% increase in successfully executed science observation targets.

1.0 The Problem Statement & Deep-Space Operational Challenges

Operating multiple scientific payloads (AFC, PALT, TIRI, HyperScout-H) in proximity to Didymos presents conflicting operational constraints:

1.1 Conflicting Payload Constraints: Running multiple high-power sensors simultaneously risks exceeding battery depth-of-discharge limits or overloading thermal radiators.
1.2 Rigid Ground Timelines: Pre-compiled ground command sequences cannot adapt to dynamic orbital perturbations or unpredicted lighting variations.
1.3 Operator Burden at ESOC: Manual ground timeline replanning consumes extensive ground controller resources.

Table 1.1: Operational Bottleneck Comparison & Quantitative Advantage

Planning Method	Adaptability to Perturbations	Payload Utilization	Overhead @ 50 MHz
Rigid Ground Uplink Timelines	None (24h turnaround)	Conservative baseline	Ground Planners
Rule-Based Event Triggers	Limited (Greedy only)	+10% efficiency gain	< 0.1 s / Sub-optimal

2.0 Mission Context & Hera Platform Technical Baseline

ARES-Planner executes within the Core 1 bare-metal sandbox, querying Data Pool parameters.

2.1 Sandbox Realism: 42.8 kB Static RAM, < 12.0 kB stack depth, zero malloc.
2.2 Mission Data Pool Inputs: Reads battery voltage (PCDU_BATT_V_VAL), MMU memory status, and orbit state.

Table 2.1: Hera Platform Allocation vs. Software Implementation Baseline

Platform Parameter	Hera Constraint	ARES-Planner Baseline	Compliance Status
Processor Architecture	50 MHz SPARC V8 (LEON3)	Pure ANSI C99 / BCC SPARC	100% Compliant
RAM Footprint	Bounded Sandbox RAM	42.8 kB Static RAM (0 malloc)	Zero Heap Used
Stack Depth	64.0 kB max	< 12.0 kB peak stack depth	+81.2% Stack Margin

3.0 Algorithmic Architecture & Software Pipeline Design

ARES-Planner deploys a 4-stage integer CSP optimization engine:

3.1 Stage 1 – Resource Envelope Ingestion: Queries telemetry to evaluate battery charge, thermal state, and MMU memory.
3.2 Stage 2 – Branch-and-Bound CSP Solver: Explores candidate activity sequences using fixed static priority trees in RAM, pruning branches that violate constraints.
3.3 Stage 3 – Master Timeline Generation: Produces a conflict-free observation schedule maximizing Science Priority Index.
3.4 Stage 4 – PUS Timeline Reporting: Emits generated schedules in PUS Science Packets (APID 0x483).

Table 3.1: Pipeline Stage Execution & Memory Budget (50 MHz SPARC V8)

Pipeline Stage	Algorithmic Function	Memory Footprint	WCET @ 50 MHz
Stage 1: State Evaluator	Battery & memory constraint boundary check	2.8 kB Scratchpad	0.05 seconds
Stage 2: CSP Solver	Branch-and-bound integer activity schedule search	28.0 kB State Tree	1.15 seconds
Stage 3: Timeline Coder	Master schedule timeline packing	8.0 kB Schedule Buf	0.15 seconds
Stage 4: PUS Reporter	PUS Science Report (APID 0x483) emission	4.0 kB Packet Buffer	0.05 seconds
TOTAL INTEGRATED PIPELINE	24-Hour Master Science Schedule Optimization	42.8 kB Static RAM	1.40 seconds

4.0 Mathematical Formulation & Implementation Equations

ARES-Planner optimizes observation schedules via integer Linear Programming / CSP formulation:

4.1 Objective Function: $\max \sum_{i=1}^N w_i x_i$ (where w_i is science priority, $x_i \in \{0, 1\}$ is activity execution flag).
4.2 Energy Constraint: $E(t) = E_0 + \int_0^t (P_{\text{solar}}(\tau) - \sum_{i} x_i P_i(\tau)) d\tau \ge E_{\min}$
4.3 Memory Ceiling: $\sum_{i} x_i M_i \le M_{\text{downlink_avail}}$
The solver uses integer arithmetic to prune invalid subtrees.

5.0 SIFT Radiation Hardening & Fault-Tolerant Execution

SIFT protections for ARES-Planner:

5.1 Schedule Integrity Voting: Activity execution flags are stored in tmr_uint32_t structures.
5.2 64-Byte Config Block (0x40001000): Ground tunable payload priority weights w_i and minimum battery reserve threshold.

Table 5.1: 64-Byte In-Flight Configurable Memory Map (Fixed Address: 0x40001000)

Offset / Field	Type	Default Value	Operational Description & Tuning Function
+0x00: config_version	uint16	0x0100	ARES-Planner configuration format identifier
+0x02: min_batt_reserve_v	uint16	2800 (28.0V)	Minimum battery bus voltage threshold
+0x04: afc_priority_weight	uint8	100	AFC optical imaging priority weight
+0x08: crc32_checksum	uint32	Computed	Configuration block integrity checksum

6.0 Platform Interface Integration & PUS Telemetry Mapping

ARES-Planner interfaces with Core 0 via standard hera_interface.h functions:

6.1 Telemetry Emission: Emits scheduled timelines in PUS Science Packets (APID 0x483) and PUS-3 Housekeeping.

Table 6.1: PUS Telemetry Packet Structures & Science Emission Budget

PUS Service / Packet	APID / SID	Cadence / Trigger	Payload Size	Telemetry Description
PUS-3 Housekeeping	APID 0x483, SID 0x0305	Every 10 minutes	128 bytes	Planner status, scheduled target count, energy reserve
PUS-20 Science Report	APID 0x483, Type 20/1	Per planning epoch	≤ 1024 bytes	Master observation activity timeline schedule

7.0 Empirical Verification & Ground Simulation Evidence

ARES-Planner was benchmarked in QEMU LEON3 using simulated Didymos orbital scenarios:

7.1 Verification Summary:
• Observation Gain: +35% increase in valid science targets scheduled.
• WCET Execution: 1.40 s per 24-hour master plan @ 50 MHz.
• Memory Allocation: 42.8 kB Static RAM, < 12.0 kB stack depth.

8.0 Operational Concept & Industrial Implementation Roadmap

8.1 Operational Timeline: Evaluates schedules at session start, outputting optimized timelines to Mass Memory.

8.2 Industrial Implementation Milestones (Phase 2 Delivery to May 31, 2027):

Milestone	Target Date	Key Deliverables & Verification Scope	Standard
MS1: Kick-Off & PDR	November 2026	Software Requirements Document (SRD), Initial Design Definition File (DDF)	ECSS Cat D
MS2: Critical Design (CDR)	February 2027	Complete C Codebase, Interface Control Document (ICD), Telemetry Maps	MISRA-C
MS3: V&V; Qualification	April 2027	QEMU Automated Test Reports, Frama-C Static Verification Proofs	ECSS V&V;
MS4: Final Flight Package	May 15, 2027	Full Source Code, DDF, SUM, ICD, R-DAS Ground Segment Decoder	Final Delivery
MS5: In-Flight Campaign	August 2027	In-Orbit Execution Support (ESOC Darmstadt Operations Team)	Mission Ops

8.3 Ground Decoder: Includes Gantt-chart timeline visualization tools for ESOC flight controllers.

9.0 Proposing Entity & Key Leadership Triad

Proposing Entity Profile: radixal s.r.o.
Established in 2016 in Brno, Czech Republic (Purkynova 649/127), radixal s.r.o. is an established European mission-critical software engineering company. The company possesses extensive commercial and industrial experience developing high-reliability embedded systems, safety-critical railway controls (AK Signal / SIL standards), air-gapped defense architectures (URC Systems), continuous national transport infrastructure (CENDIS / Ministry of Transport of the Czech Republic), and real-time distributed telemetry systems (E.ON, Schneider Electric, Swiss Life Select).
• Proven European Spaceflight & Satellite Imagery Heritage (Spacemetric AB / Norway & Sweden): Direct engineering partnership on native C/C++ image decompression and high-performance processing engines for Spacemetric (repo: gitlab.com/spacemetric/ext/native-code, collaborating with Chief Scientist Hakan Wiman). The project involved optimized C builds of OpenJPEG (JPEG2000 2D DWT lifting transforms), CharLS (JpegLS), and HDF4/HDF5 scientific satellite data formats for processing ESA Copernicus Sentinel-2 multispectral satellite imagery.

Key Leadership Triad:
• Bc. Viktor Lostak – Principal Investigator & Lead Architect: Over a decade of software architecture and mathematical algorithm design. Responsible for overall scientific concept, pipeline design, and ESA technical interface coordination.
• Ing. Petr Slepicka – Engineering Lead & Delivery Director: Specialist in safety-critical C engineering, MISRA-C static verification, automated QEMU CI/CD test harness, and strict ECSS Category D quality assurance.
• Mgr. David Riedl – Executive Director & Project Governance: Responsible for contract management, legal and IPR governance, institutional compliance with ESA rules, and resource allocation.

10.0 Scientific References & Proposed External Advisory Board

Academic Citations & Conceptual Foundation:
[1] Chien, S., et al., „Activity-Based Operations: Integrating Planning and Scheduling in Spacecraft Autonomy“, IEEE Aerospace.
[2] Carnelli, I., et al., „The ESA Hera Mission: Detailed Characterization“, Advances in Space Research, 2022.
[3] ECSS Secretariat, „ECSS-E-ST-40C: Space engineering – Software“, ESA-ESTEC, 2020.

Proposed External Advisory & Review Board:
radixal s.r.o. formally proposes establishing an External Advisory Board inviting technical consultations with the ESTEC Flight Software Systems Section (TEC-SW) and the Astronomical Institute of the Czech Academy of Sciences (Ondrejov Observatory), leading to joint peer-reviewed paper publication at the DASIA 2028 and EDHPC 2028 conferences.